

International Capital Flows in the Context of Financial Stability

Financial Stability

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- 2 The Intertemporal Approach to the Current Account
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- 7 Geopolitics, Public Debt, and De-Dollarization
- 8 Conclusions

Balance of Payments (BoP) Framework (BPM6)

The Fundamental Identity

The BoP is a double-entry accounting tautology capturing all economic transactions between residents and non-residents:

$$CA + KA + NEO = FA \quad (\text{or } CA + KA + NEO = \Delta NFA)$$

A current account deficit is not “lost trade”; it is fundamentally a reflection of net capital inflows. Every transaction is an asset exchange.

- **Current Account (CA):** Goods, services, primary income (returns on investment), secondary income.
- **Capital Account (KA):** Non-produced, non-financial assets (negligible for advanced economies).
- **Financial Account (FA):** FDI, Portfolio Investment, Other Investment (bank loans, deposits), Reserve Assets.

Sectoral Balances and the Macroeconomic Identity

From the national income identity $Y = C + I + G + (X - M)$, we derive the Sectoral Financial Balances:

$$(S_p - I_p) + (T - G) = (X - M) = CA$$

Where:

- $(S_p - I_p)$ is the **Private Sector Balance**.
- $(T - G)$ is the **Public Sector Balance** (Government Surplus/Deficit).
- $(X - M) \approx CA$ is the **Foreign Sector Balance**.

Core Implication: A Current Account deficit ($CA < 0$) *must* be offset by either a private sector deficit (e.g., US household borrowing pre-2008) or a public sector deficit. It is fundamentally a saving-investment gap, not just a trade phenomenon.

The Intertemporal Approach: Core Theory

The Intertemporal Approach (IA) views the Current Account as the optimal, forward-looking outcome of consumption smoothing and investment decisions by rational agents.

Intertemporal Budget Constraint (IBC): A nation cannot run Ponzi schemes; the present value of future trade surpluses must equal current net foreign debt (B_t).

$$-(1+r)B_{t-1} = \sum_{s=t}^{\infty} \frac{TB_s}{(1+r)^{s-t}}$$

Consumption Smoothing:

- **Temporary Shocks:** A temporary drop in income (e.g., natural disaster) leads to a CA deficit as the country borrows to smooth consumption.
- **Permanent Shocks:** A permanent drop in income leads to an immediate adjustment in consumption, leaving the CA largely unchanged.

Two-Period Small Open Economy Model

Assumptions: Two periods, single good, exogenous r^* , endowments Y_1, Y_2 , $U = u(C_1) + \beta u(C_2)$, $B_0 = B_2 = 0$.

Budget constraints:

$$C_1 + B_1 = Y_1, \quad C_2 = (1 + r)B_1 + Y_2$$

Intertemporal BC:

$$C_1 + \frac{C_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r}$$

Utility maximization $\Rightarrow$ Euler equation:

$$\frac{u'(C_1)}{u'(C_2)} = \beta(1 + r)$$

Log utility example ($\beta = 1/(1 + r)$ for perfect smoothing):

$$C_1 = C_2 = \frac{(1 + r)Y_1 + Y_2}{2 + r}$$

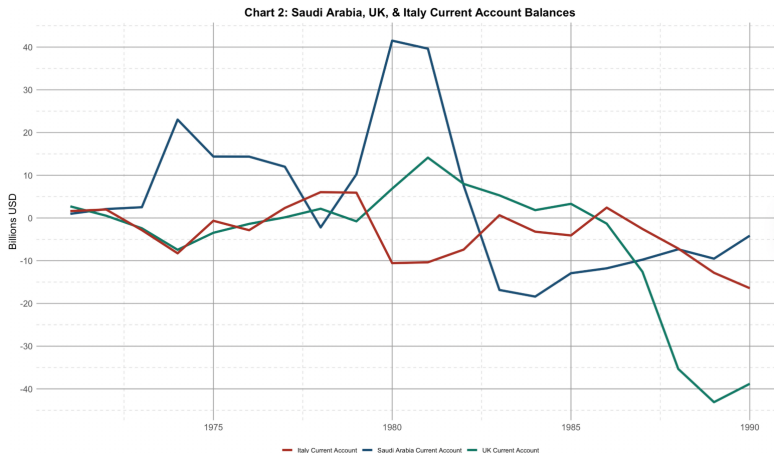
$CA_1 = Y_1 - C_1 = B_1$ (surplus if Y_1 is temporarily high),

Applications: Oil Shocks, Immigration, and Debt Crises

- **1970s Oil Crises:** Importers ran deficits by borrowing to smooth consumption; exporters ran surpluses by lending.
- **Israel's 1990s Immigration:** Temporary income drop led to CA deficit for smoothing.
- **U.S.-China Imbalances:** U.S. deficits from low saving; China surpluses from high growth expectations.
- **Sovereign Debt:** Unsustainable if required surpluses exceed feasible NX; e.g., Greece needed surplus but ran deficits.

Intertemporal Approach Example

Oil Price Shocks of the 1970s



Source: <https://gianlucabenigno.substack.com/p/background-note-on-trade-and-trade>

Current Account Surpluses in Developed Economies

Structural ("natural") drivers:

- Aging demographics $\Rightarrow$ high private saving.
- Positive net international investment position ("renter" status).
- High capital/labor ratio $\Rightarrow$ diminishing returns to investment.
- Technological leadership + high value-added exports.
- Low investment rates relative to saving.

These create persistent surpluses without policy intervention.

Lucas Paradox: Deficits in Developed Economies

Capital flows “uphill” from poor to rich (contrary to neoclassical prediction of high returns in capital-scarce EMs).

Possible Drivers:

- Falling household saving + consumption excess productivity growth.
- Low real rates + rising household/public debt.
- Boom in non-productive investment (housing, real estate).
- Labor market rigidities + rising unit labor costs.
- Loss of competitiveness (asymmetric trade barriers).

From Theory to Crisis: What the IA Misses

So far: The IA elegantly explains why countries run surpluses or deficits as rational, forward-looking behaviour.

The problem: It models *net* resource transfers. While persistent CA deficits signal a **medium/long-term vulnerability** (structural fragility), they are rarely the immediate cause of a crisis.

Modern financial crises are almost universally triggered by **sudden stops in gross financial flows**—bank lending, derivatives, wholesale funding—dynamics that the CA balance fails to capture.

The Breakdown of the Intertemporal Approach

While the IA elegantly models **real resource transfers** (Net Flows), it fails to capture the mechanics of modern financial crises.

Critiques by Borio, Disyatat, and Shin:

- **Saving is not Financing:** The IA conflates Saving (an ex-post national accounting residual representing foregone consumption) with Financing (an ex-ante cash-flow concept providing purchasing power).
- **Endogenous Money:** Banks do not merely intermediate pre-existing savings (Loanable Funds theory). Banks create purchasing power ex nihilo when they extend credit. Investment creates its own financing, decoupling credit booms from the Current Account.
- **The "Excess Elasticity" of Credit:** Financial fragility stems from the financial system's ability to expand balance sheets infinitely, not from the "Global Saving Glut".

Gross vs. Net Capital Flows: The Vulnerability-Trigger Nexus

The Current Account (Net Flow) acts as a slow-moving vulnerability indicator, but masks the massive, highly volatile **Gross Flows** that act as the actual crisis trigger.

The 2008 Financial Crisis Example

Prior to 2008, the net CA between Europe and the US was relatively balanced. However, **Gross Flows** exploded:

- European banks borrowed heavily in US wholesale money markets (increasing US foreign liabilities).
- European banks used those USD funds to buy US Mortgage-Backed Securities (increasing US foreign assets).

The Result: Net flows were zero, but gross exposures were enormous. When the US housing market turned, European banks faced a massive **Sudden Stop** in USD funding, triggering a global crisis. The CA showed no vulnerability; gross flows showed an imminent collapse.

What the Data Tell Us: A Road Map

Four empirical regularities to keep in mind

What we are looking for

Theory predicts that capital should flow to where returns are highest. The data show a more complex picture.

- 1 **FDI is the anchor.** It is the least volatile component of capital flows and the least linked to sudden stops.
- 2 **Global imbalances are structural, not temporary.** The US has run a CA deficit for four decades; Asia has run surpluses for the same period.
- 3 **Saving rates explain the surplus side.** East Asia saves 35–50% of income; North America saves under 20%. That gap funds US deficits.
- 4 **Valuation effects matter as much as flows.** The US NIIP is far less negative than cumulative CA deficits imply — exchange rate and equity price effects distort the headline number.

Empirical Patterns: FDI Inflows (1990–2024)

By Development Level

World FDI dominated by developed economies until 2005; developing economies (esp. Asia) surged post-2010.

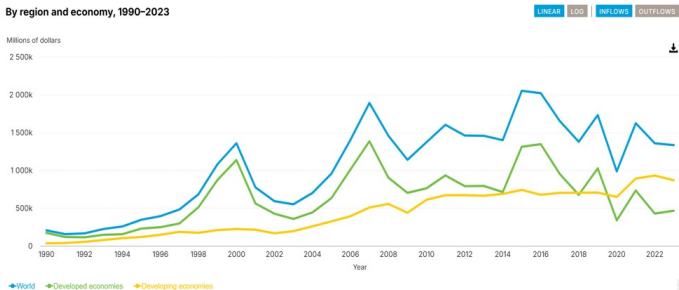
By Region

- Asia: Explosive growth (China + India).
- Europe/North America: Volatile leaders.
- Africa/Latin America/Caribbean: Smaller, commodity-driven.

FDI is the *least volatile* component of capital flows (BPM6). Portfolio and “other” flows (wholesale banking) dominate volatility and crisis transmission.

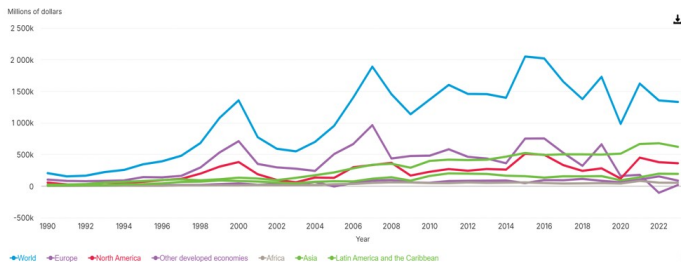
FDI Inflows (1)

By region and economy, 1990–2023



FDI Inflows (2)

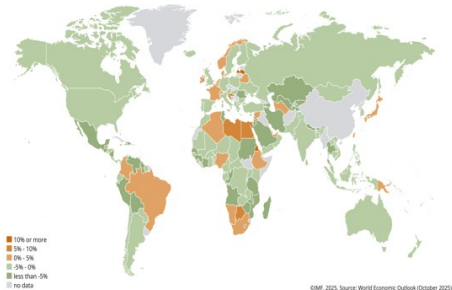
By region and economy, 1990–2023



The Current CA Imbalances (World)

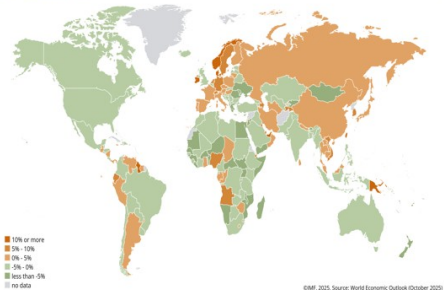
IMF DataMapper

Current account balance, percent of GDP (Percent of GDP, 1992)



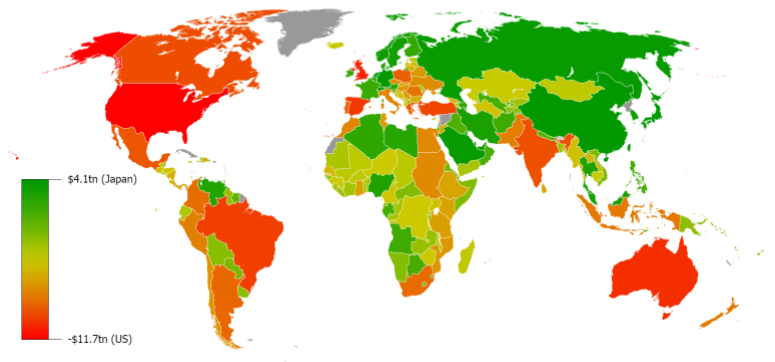
IMF DataMapper

Current account balance, percent of GDP (Percent of GDP, 2024)



Source: *Dan Wales Lecture notes*

Cumulative Current Account Balances



Source: Dan Wales Lecture notes

The Persistent U.S. Current Account Deficit

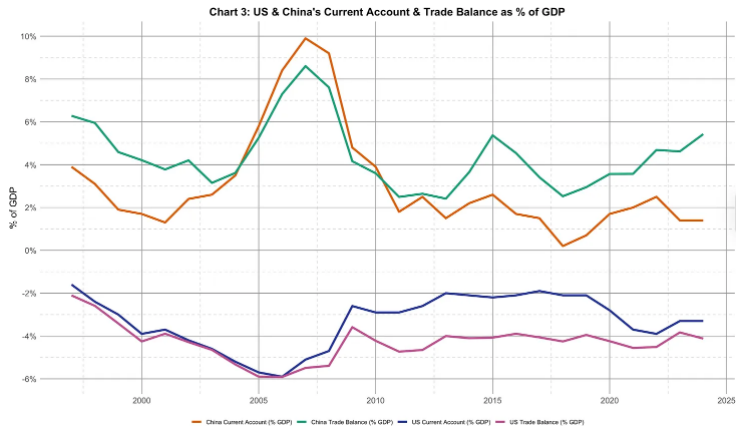
- High deficit since mid-1980s, accelerated 1990s–pre-2008.
- Financed by foreign capital (zero reserve change) thanks to USD reserve status.
- Consumption/investment boom in non-tradables + export sector decline.
- Asset bubbles fueled by inflows.

Supporting evidence:

- CA and income balance (% GDP).
- Private saving/investment + government saving.
- Net private + household saving rates (% disposable income).

Deficits reflect saving-investment gaps, not “unfair trade.”

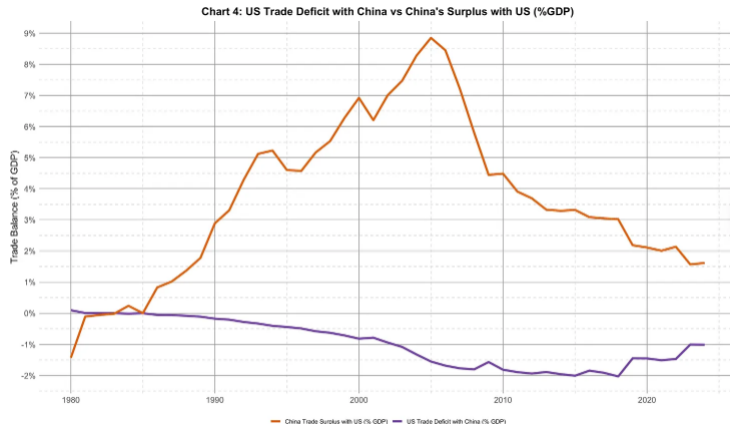
The Current Imbalances (US and China)



Source: <https://gianlucabenigno.substack.com/p/background-note-on-trade-and-trade>

The Current Imbalances (US and China)

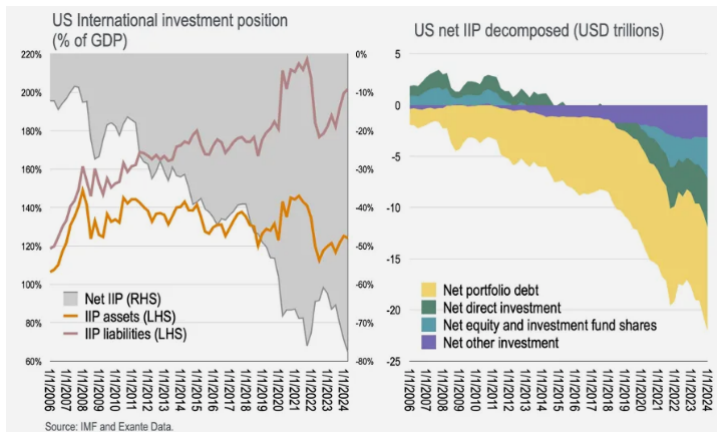
A Closer Look



Source: <https://gianlucabenigno.substack.com/p/background-note-on-trade-and-trade>

US International Investment Position

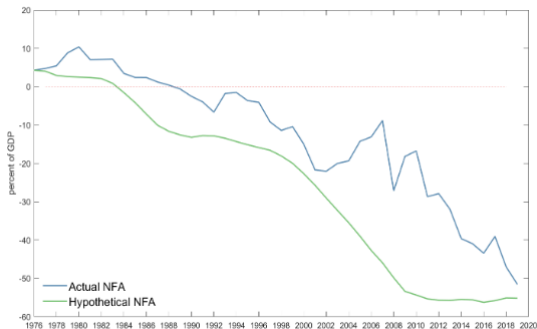
Exorbitant Privilege?



Source: <https://moneyinsideout.substack.com/p/united-states-changing-net-iip>

The US NIIP and Valuation Changes

Hypothetical NIIP Without Valuation Changes Since 1976



Source: https://people.brandeis.edu/~ghall/econ303/build/html/niip_paradox.html

NIIP: Beyond Trade Balances

The Net International Investment Position (NIIP) is the value of U.S.-owned foreign assets minus foreign-owned U.S. assets.

The NIIP Identity:

$$\Delta NIIP_t = CA_t + \Delta VA_t + RES_t$$

Where CA is the Current Account, and ΔVA represents **Valuation Changes**.

- **Price Effects:** Changes in the market value of stocks and bonds.
 - If the U.S. stock market outperforms foreign markets, U.S. *liabilities* rise faster than assets, making the NIIP more negative.
- **Exchange Rate Effects ("Exorbitant Privilege"):**
 - U.S. liabilities are mostly in USD; U.S. foreign assets are often in foreign currencies.
 - **A weakening Dollar** increases the USD-value of foreign assets, mechanically improving the U.S. NIIP.

Key Takeaway: In the 2020s, valuation effects (growth in U.S. equity values) have often overshadowed the trade deficit (CA) in driving the NIIP's trajectory.

Asian Economies Financing Global Deficits

China and Asian EMs:

- High private saving + export-led growth (productivity surges).
- Massive reserve accumulation (self-insurance + competitiveness via pegs).
- Public (not private) financing of developed-world deficits.
- Young demographics initially boosted saving.

Consequences (Lucas Paradox in reverse):

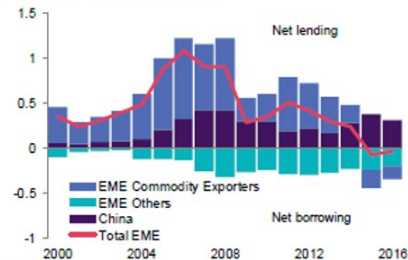
- U.S. debt financed by EM reserves.
- Shift in global asset ownership toward EMs.

Asian Economies Financing Global Deficits

22. Current account breakdown across EM

Source: NWM, BoE

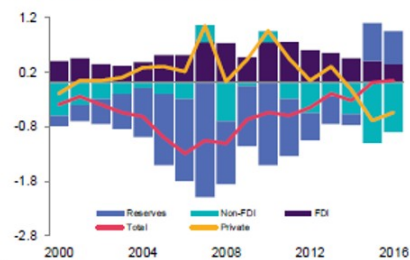
Current account as % of world GDP



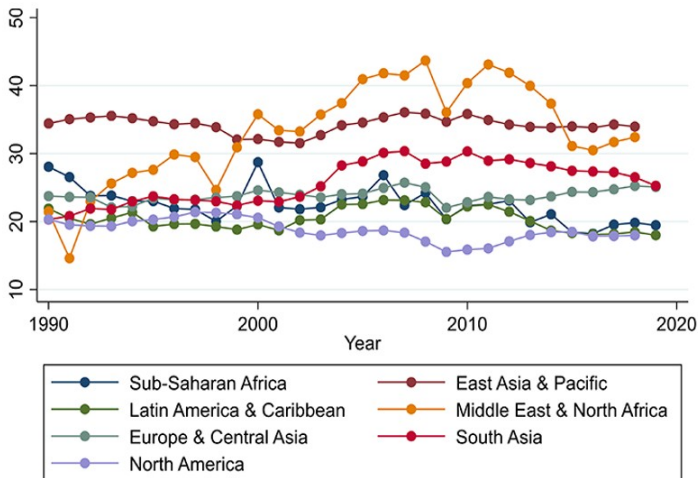
23. Type of capital flows to EM

Source: NWM, BoE

Net capital flow to EM as % of world GDP



Saving Rates Comparison



US Saving Rate



The Kalecki Profit Equation in an Open Economy

Michal Kalecki demonstrated that macroeconomic aggregates dictate corporate profits. Assuming workers consume what they earn ($C_w = W$), aggregate profits (P) are defined as:

$$P = I + (G - T) + (X - M) + C_k - S_w$$

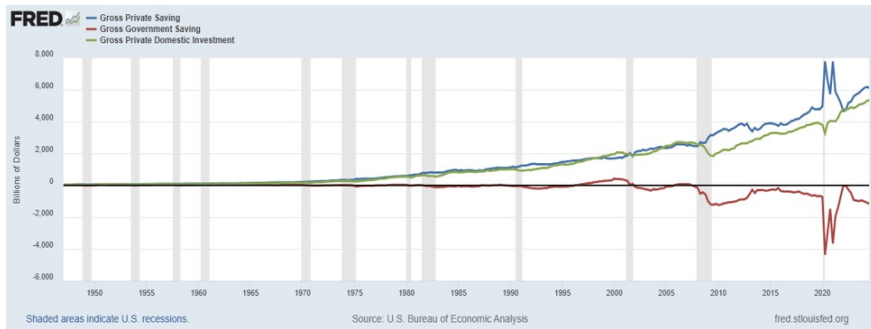
Simplified, Corporate Profits equal:

Profits = Investment + Government Deficit + Current Account Surplus + Capitalist Consumption

Application to Global Flows:

- **Export-Led Economies (Germany, China):** Rely heavily on the $(X - M)$ term (CA surpluses) to sustain high corporate profitability without requiring domestic public/private deficits.
- **Deficit Economies (US, UK):** Because $(X - M)$ is negative, corporate profits *must* be sustained by massive Government Deficits $(G - T)$ or debt-fueled domestic Investment/Consumption.

Gross Private Savings, Investment, and Government Savings



From Global Patterns to a Case Study

Pattern established: Gross flows dominate volatility; FDI is stable, portfolio and banking flows are not.

Next: The Eurozone (1999–2010) is the clearest case where a sudden stop in gross banking flows — acting on the medium-term vulnerability of divergent ULCs — triggered a systemic crisis.

The Eurozone Debt Crisis: Gross Flows and RER

The Eurozone provides a textbook case of capital flow misallocation within an incomplete Optimum Currency Area (OCA).

The Setup (1999–2007):

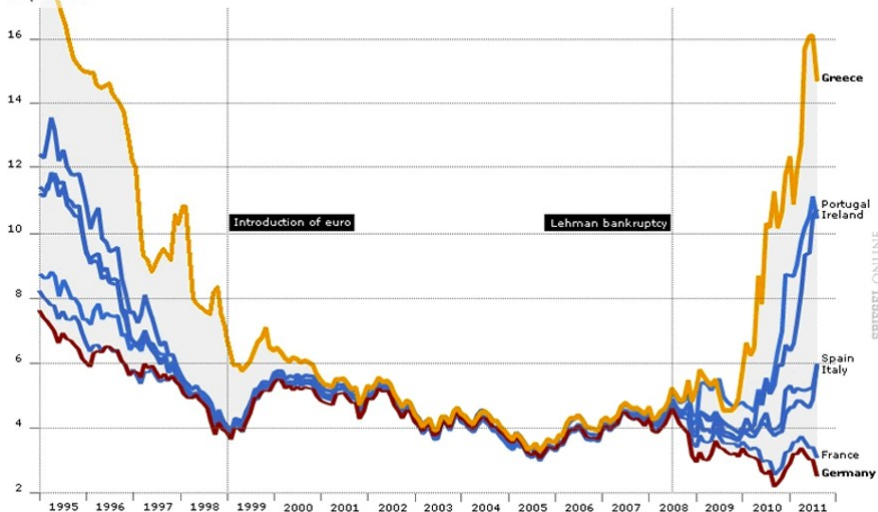
- Monetary union eliminated nominal exchange rate risk and compressed sovereign spreads:

$$i^{\text{Periphery}} \downarrow \rightarrow i^{\text{Periphery}} \approx i^{\text{Core}}$$

- Large **gross capital flows** from Core (e.g., Germany) to Periphery (e.g., Spain, Greece), intermediated via the banking system.
- Capital inflows financed domestic demand booms:
 - Spain: Housing and construction boom.
 - Greece: Public consumption and fiscal expansion.
- Weak financial intermediation → misallocation toward non-tradables.

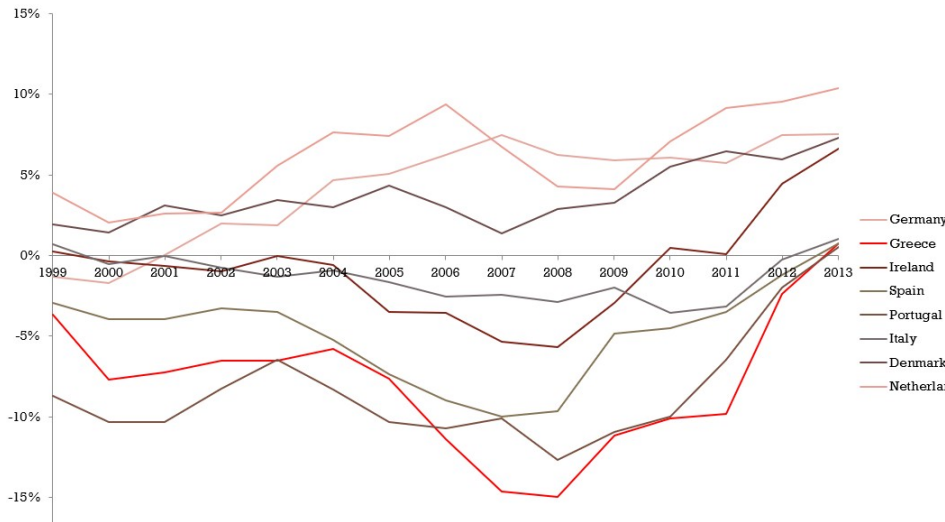
Yields Comparison

Development in Interest Rates on 10-year Government Bonds
in percent

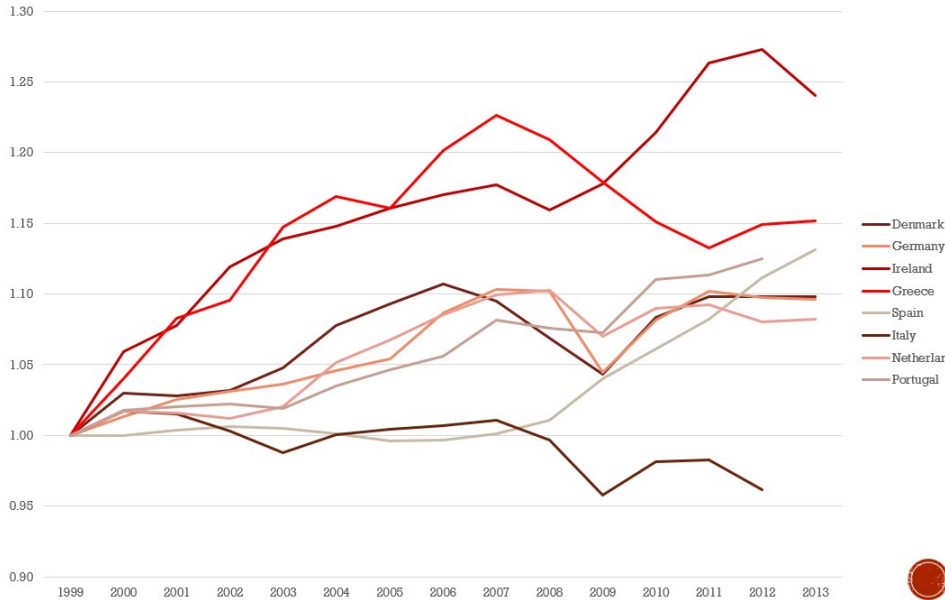


Source: Thomson Reuters Datastream

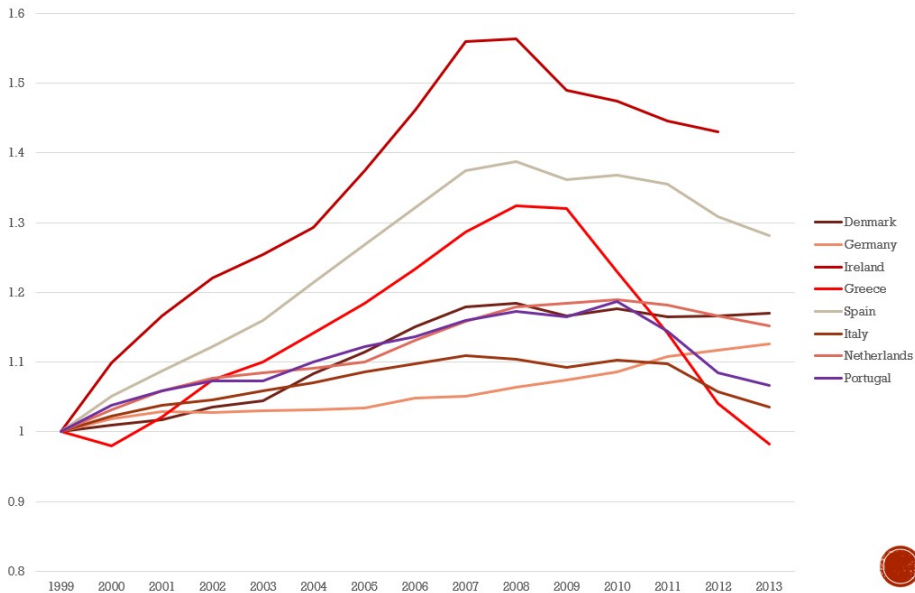
Current Account Comparison



Productivity Comparison



Consumption (Private + Government) Comparison



Unit Labor Costs (ULC) and Cost Pressures

Define Nominal Unit Labor Cost (w = nominal wage, L = labor, Y = real output):

$$ULC_n = \frac{wL}{Y} = \frac{w}{Y/L}$$

- ULC_n increases when nominal wage growth (Δw) exceeds productivity growth (Δa):

$$\Delta ulc_n \approx \Delta w - \Delta a$$

- **In the Periphery:** Large capital inflows drove up wages without matching productivity gains.

$$\Delta w \gg \Delta a \Rightarrow ULC_n \uparrow$$

- **In the Core (e.g., Germany):** Wage moderation kept costs in check.

$$\Delta w \approx \Delta a \Rightarrow ULC_n \text{ stable}$$

Real Unit Labor Costs and the Labor Share

To find Real ULC (ULC_r), we deflate Nominal ULC by the price level (P):

$$ULC_r = \frac{ULC_n}{P} = \frac{wL}{PY} = \frac{w/P}{Y/L}$$

- Real ULC measures the ratio of the **real wage** (w/P) to **labor productivity** (Y/L).
- Economically, it represents the **labor share of income**—the proportion of national income paid to workers.
- In growth rates:

$$\Delta ulc_r \approx \Delta w_{real} - \Delta a$$

- **Takeaway for the Eurozone Crisis:** While Nominal ULC measured the loss of export competitiveness, Real ULC tracked whether workers or capital owners captured the gains of the boom.

Real Exchange Rate (REER) Channel

With fixed nominal exchange rates in the Eurozone, cost competitiveness is determined by relative Nominal ULCs:

$$REER \sim \frac{ULC^{\text{Periphery}}}{ULC^{\text{Core}}}$$

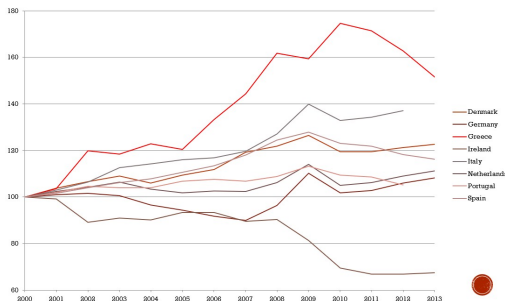
- Rising relative ULC → **real appreciation**.
- Loss of external competitiveness (exports ↓, imports ↑).
- Growing current account deficits financed by gross inflows.

Crisis Dynamics (Post-2010):

- Sudden Stop in gross capital flows → funding crisis in banking systems.
- No nominal devaluation possible (monetary union constraint).
- Adjustment via **internal devaluation**: wage compression, high unemployment, and severe economic contraction.

ULC Divergence: Core vs Periphery (Nominal)

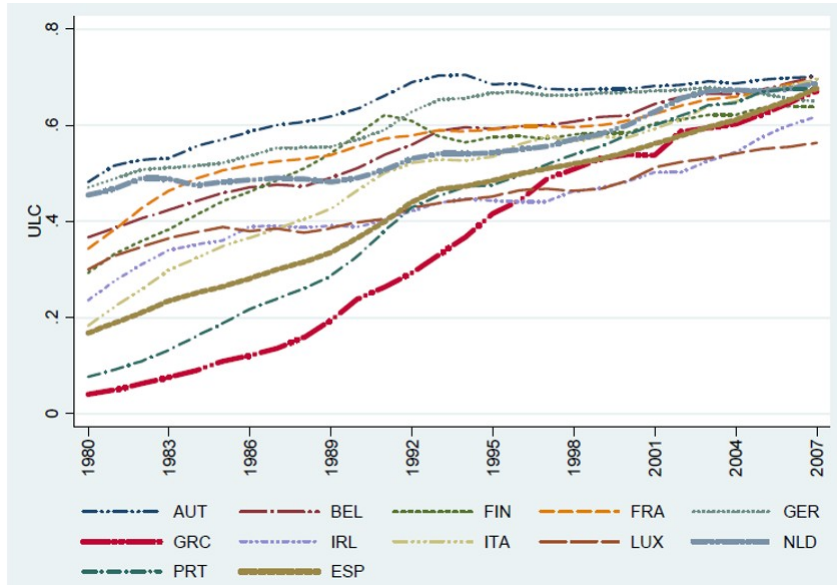
What the data show



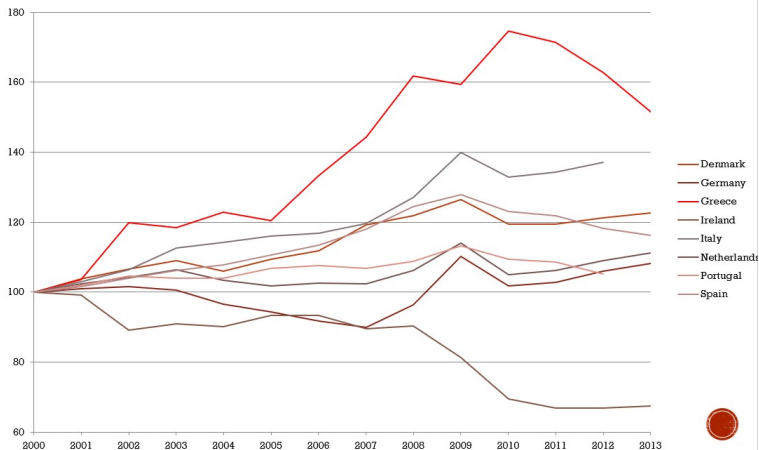
Read-off:

- Greece nominal ULC rose ~75% (2000–2008); Germany flat.
- Mechanism: capital inflows $\Rightarrow$ wage growth $\gg$ productivity growth.
- No devaluation possible $\Rightarrow$ competitiveness loss baked in.

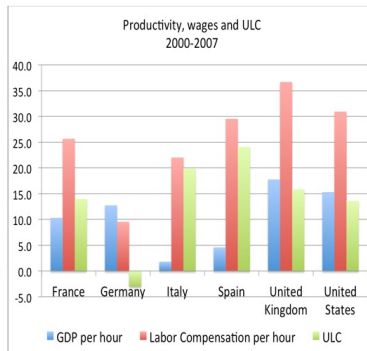
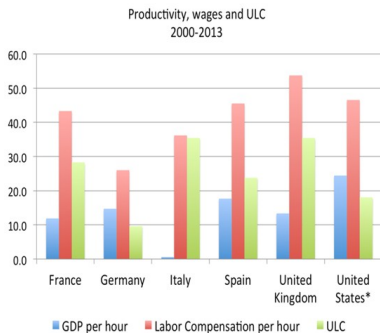
ULC Divergence: Core vs Periphery (Nominal)



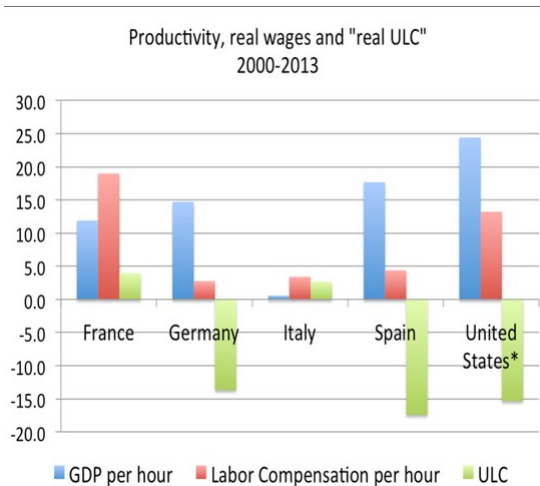
Unit Labor Cost - Comparison 1 (Nominal)



Unit Labor Cost - Comparison 2 (Nominal Wages)



Unit Labor Cost - Comparison 3 (Real Wages)



The "Competitiveness Paradox" (2000–2007)

During the boom years in the Periphery, a striking paradox emerged:
Nominal ULC surged while Real ULC remained flat or fell.

The Mechanism:

- Capital inflows drove massive domestic demand → high non-tradable inflation ($P \uparrow$).
- Tight labor markets pushed up nominal wages ($\Delta w > 0$).
- Because productivity stagnated ($\Delta a \approx 0$), **Nominal ULC surged** ($ULC_n = w/(Y/L)$).

Why Real ULC Fell ($ULC_r = ULC_n/P$):

- Firms had immense pricing power during the boom. They raised prices (P) faster than they raised nominal wages (w).
- Since P grew faster than ULC_n , the Real ULC actually **declined**.

The Fatal Policy Error:

- Looking only at Real ULC (labor share of income) suggested labor was getting "cheaper" relative to output.
- But the surge in Nominal ULC was actively pricing these countries out of global markets, building massive external imbalances.

The Eurozone Policy Response (2010–2015)

What Was Done — and What Remains Unresolved

Crisis Management (short-run firefighting):

- **European Financial Stability Facility (EFSF) / ESM (2010/2012):** Bailout funds providing conditional loans to Greece, Ireland, Portugal, Spain. Conditionality tied to fiscal austerity and structural reforms.
- **ECB: Securities Markets Programme (SMP, 2010):** Sterilised secondary-market purchases of peripheral sovereign bonds.
- **ECB: Outright Monetary Transactions (OMT, 2012):** Draghi's “whatever it takes” backstop. *Unlimited*, conditional sovereign bond purchases.

Structural Reforms (medium-run):

- **Banking Union (2014):** Single Supervisory Mechanism (SSM) at the ECB; Single Resolution Mechanism (SRM). Broke the *bank-sovereign doom loop* by moving supervision above the national level.
- **Fiscal Compact (2012):** Treaty-level structural deficit limits; automatic correction mechanisms.

The Eurozone Policy Response — What Remains Unresolved

Incomplete Architecture

The Eurozone fixed the acute crisis but left the root structural flaws largely intact.

- **No European Deposit Insurance Scheme (EDIS):** National deposit guarantee schemes remain, preserving bank-sovereign linkage at the margin. Germany blocks mutualisation.
- **No Fiscal Union / Eurobonds:** The EU lacks a common safe asset that would recycle surplus savings toward deficit regions automatically (à la US Treasuries).

Takeaway: The Eurozone response *stabilised* the crisis but did *not* resolve the fundamental tension between a monetary union and independent fiscal policies with divergent competitiveness.

Scaling Up: From Eurozone to Global Dollar Risk

Eurozone lesson: Sudden stops in gross flows can force painful internal adjustment with no exchange rate valve. **The same risk exists globally:** The US runs persistent deficits financed by foreign gross inflows. What happens if those flows slow or reverse?

Reserve Currencies: The Basics

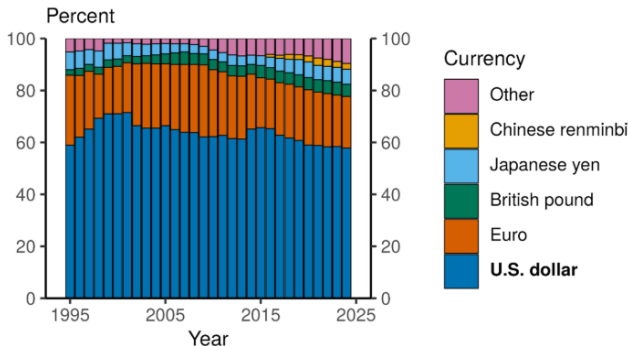
- **Definition:** Foreign currencies held by central banks for international transactions, stability, and reserves.
- **Dominance:** Driven by the issuer's economic size, stability, and **network effects** (inertia).
- **Key Characteristics:**
 - **Liquidity:** Deep, accessible financial markets.
 - **Stability:** Low inflation/default risk.
 - **Convertibility:** Unrestricted capital flows.
 - **Institutional Strength:** Rule of law, central bank independence.

Current Global Foreign Exchange Reserves

Global Holdings Remain Dominated by USD and EUR

Reserve currencies are held worldwide as safe, liquid assets.

- **USD Dominance:** Dominance persists despite a decline from over 70% in the early 2000s.
- **Renminbi (CNY):** Limited by capital controls and credibility concerns, restricting its growth.



Source: <https://www.federalreserve.gov/econres/notes/feds-notes/>

Current Status of Reserve Currencies

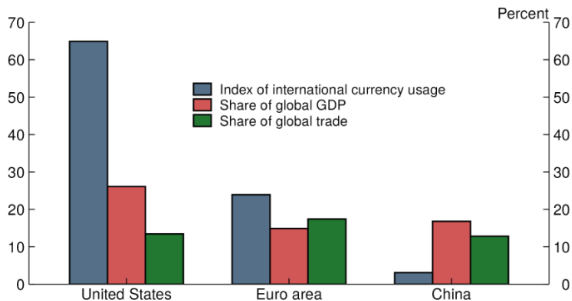
Based on IMF COFER data (as of Q2 2025):

Currency	Share (%)
US Dollar (USD)	58.0
Euro (EUR)	20.0
Japanese Yen (JPY)	5.5
British Pound (GBP)	4.8
Chinese Renminbi (CNY)	2.8
Canadian Dollar (CAD)	2.5
Australian Dollar (AUD)	2.0
Others	4.4

Table: Approximate Global Reserve Shares (Q2 2025)

Note: USD share has declined from over 70% in the 2000s due to diversification and geopolitical shifts.

International Role vs. Size of Economy (2024)



Source: <https://www.federalreserve.gov/econres/notes/feds-notes/the-international-role-of-the-u-s-dollar-2025-edition-20250718.html>

The US Dollar's Multifaceted Dominance

Why the USD remains the 'World's Currency'

- **Reserve Share:** 58% of global reserves (2024).
- **Trade and FX:** $\approx 88\%$ of all FX trades involve USD (BIS 2022); 96% of Americas invoicing.
- **Payment Systems:** $\approx 50\%$ of global SWIFT payments.
- **The Core Support:**
 - Deepest, most liquid **Treasury Market** (Safe Assets).
 - Support from **Fed Swap Lines** (critical backstop liquidity).
 - Strong institutional framework (**Rule of Law**).

Geopolitics and Shifting Capital Flows

The global consensus around the USD as the ultimate safe haven is fracturing due to weaponization (sanctions) and US fiscal dominance.

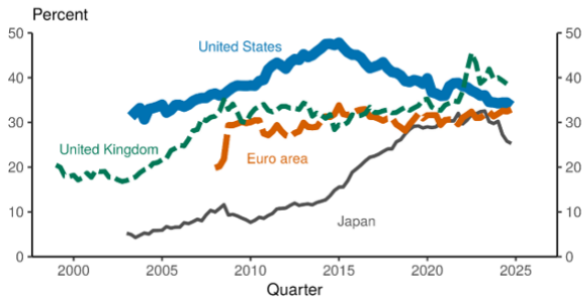
The Changing Composition of Global Financing:

- **Declining Foreign Demand for US Debt:** Foreign holdings of US Treasuries have fallen from nearly 50% in 2013 to ~ 31% today.
- **The Rise of Gold:** For the first time, the value of physical gold in global central bank reserves has surpassed the value of their US Treasury holdings.
- **Fiscal Dominance Risks:** With US debt servicing costs exceeding military spending, markets increasingly fear Federal Reserve independence will be compromised to monetize the debt.

Implication: Global capital flows are fracturing into multipolar blocs. Deficit countries (like the US) can no longer take foreign financing of their deficits for granted.

Securities - Foreign Share

4b. Share of general government debt securities held by foreign investors



Source: <https://www.federalreserve.gov/econres/notes/feds-notes/the-international-role-of-the-u-s-dollar-2025-edition-20250718.html>

The Rise of Gold



Conclusions (I): Vulnerabilities vs. Triggers

- **Accounting Realities (The Vulnerability):** Current Account deficits reflect structural saving-investment imbalances $((S - I) + (T - G))$. They act as a *medium-to-long-term vulnerability gauge* rather than an immediate crisis trigger.
- **Look Beyond the Net:** Financial crises are rarely caused by Net trade imbalances. They are fueled by the buildup of **Gross Capital Flows** (bank lending, derivatives) generated by endogenous credit creation.
- **Sudden Stops (The Trigger):** Crises are ignited by abrupt reversals in these highly volatile gross flows. These sudden stops create severe, immediate liquidity and solvency crises that expose the underlying structural vulnerabilities.

Conclusions (II): Global Distortions and Tail-Risks

- **Asset Market Distortions:** Persistent imbalances sustain bubbles and non-productive investment. Furthermore, the modern shift toward equity portfolio flows heavily skews global valuations far beyond domestic fundamentals.
- **Structural Fragility & Leverage:** The dependence of U.S. debt financing on foreign capital exposes systemic fragility, granting political leverage to reserve-accumulating nations (e.g., via diversification away from Treasuries).
- **The Geopolitical Limit:** The exorbitant privilege of the US dollar is being tested. Unchecked fiscal (twin) deficits in a shifting, multipolar world raise severe tail-risks for global financial stability if foreign gross inflows suddenly stop.

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